

Title: Illusory generalizability of clinical prediction models

Aim

The primary aim of the research was to scrutinize the optimism surrounding the use of machine learning models to improve medical decision-making by rigorously examining their generalizability across independent clinical trials.

Using antipsychotic treatments for schizophrenia as a case study, the research specifically aimed to:

- Evaluate Out-of-Sample Performance: The study directly compared how a model performed on its initial training sample versus its performance on truly independent clinical trial samples.
- Identify Key Risks: Researchers sought to assess two major risks to model effectiveness:
 - Overfitting: Where a model fits the "random noise" of a specific dataset rather than a true signal that would generalize across different samples.
 - Poor Transportability: Where models lack external validity because patient characteristics, providers, or implementation methods vary across different trials.
- Assess Clinical Utility: The ultimate goal was to determine if these models can robustly predict outcomes for unseen, future patients, which is an essential step for precision medicine to actually improve clinical practice.

The researchers utilized data from five international, multisite randomized controlled trials (RCTs) obtained through the Yale Open Data Access (YODA) Project to conduct this multi-scenario evaluation.

Theoretical Framework / Key Concepts

The research is grounded in the framework of precision medicine, which aims to use machine learning to identify models that predict how individual patients will respond to specific treatments. The study specifically interrogates the generalizability of these models - their ability to robustly predict outcomes for unseen, future patients - using antipsychotic treatments for schizophrenia as a case study.

Core Theoretical Framework

The study is built upon a multi-scenario validation framework designed to test the limits of clinical prediction models across five independent, multisite randomized controlled trials (RCTs). This framework compares model performance across four distinct conditions:

- Within-trial (No Validation): Evaluating a model on the exact same sample used to develop it.
- Within-trial (Cross-Validation): Using techniques like k-fold cross-validation to estimate accuracy by splitting a single dataset into training and testing subsets.
- Paired-trial Validation: Training a model on one independent trial and testing it on another.
- Leave-one-trial-out Validation: Training on an aggregate of multiple trials and testing on a single excluded trial to see if increased data volume improves generalizability.

Key Concepts

- Generalizability and External Validity: These terms refer to the capacity of a model to remain accurate when applied to different patient populations, providers, or clinical settings. The research suggests that current models often have "illusory" generalizability, performing well internally but failing externally.
- Overfitting: This is a primary risk where a model fits the "random noise" of a specific dataset rather than a true biological or clinical signal. Overfitted models produce high accuracy in training data but fail to generalize to new patients.
- Transportability: This concept addresses whether a model can be moved (transported) from one context to another. Poor transportability occurs when variations in implementation or patient characteristics across trials render a model ineffective in a new setting.

- Context-Dependence: The research concludes that treatment outcome models in schizophrenia are "highly context-dependent," meaning their predictive power is often limited to the specific clinical trial environment in which they were created.
- Balanced Accuracy: A statistical metric calculated as $(\text{sensitivity} + \text{specificity}) / 2$. It was used as the primary measure of performance because its null distribution is centered at 50%, providing a clearer view of whether a model is performing better than random chance.
- Clinical Heterogeneity: The idea that schizophrenia is a broad diagnostic category containing patients at different disease stages and with varying symptoms, which makes universal outcome prediction difficult.

Methodological Concepts

- Elastic Net Algorithm: A penalized regression method that balances "stability with parsimony". It is particularly useful for medical data where predictors (like blood chemistry or psychiatric history) may be highly correlated.
- PANSS (Positive and Negative Syndrome Scale): The standardized clinical scale used across all trials to measure the severity of schizophrenia symptoms and determine treatment response.
- RSWG Criteria: Defined by the Remission in Schizophrenia Working Group, this served as the primary clinical outcome to define whether a patient had achieved remission.

Assumptions

The research into the generalizability of clinical prediction models for schizophrenia relies on several statistical, methodological, and clinical assumptions:

Statistical and Data Assumptions

- Missing Data Mechanism: The researchers assumed that missing data for certain clinical tests (like urinalysis or blood draws) were missing completely at random (MCAR). This was supported by a secondary check showing that missingness was not significantly associated with the primary treatment outcome.
- Imputation Accuracy: Based on the MCAR assumption and the low overall level of missingness (typically 0% to 8%), the authors assumed that imputing missing values with median values would not meaningfully bias the prediction models.
- Predictive Thresholds: In calculating sensitivity and specificity, the researchers assumed a 50% predictive threshold. This assumes that sensitivity and specificity are of equal clinical importance, as they had no reason to weight one more heavily than the other.
- Metric Selection: The use of balanced accuracy is based on the assumption that its null distribution is centered at 50%, allowing for a clear comparison against random chance.
- Algorithm Suitability: The choice of the elastic net algorithm assumes that predictors may be sparsely endorsed and that many covariates are likely correlated with one another.

Clinical and Theoretical Assumptions

- Standardisation of Dosage: The researchers assumed that different antipsychotic drug doses could be accurately compared by standardising them to paliperidone dose equivalents using the defined daily dose method.
- Primary Outcome Validity: The study assumes that the Remission in Schizophrenia Working Group (RSWG) criteria is the most appropriate primary metric for treatment response because it is the most commonly accepted standard in the field.
- Representative Case Study: A core assumption of the research is that antipsychotic treatment for schizophrenia serves as a valid and informative case study for examining the broader challenges of machine learning generalizability in medicine.

- Requirement for Clinical Utility: The theoretical framework of the paper assumes that for precision medicine to be effective, models must robustly predict outcomes for unseen, future patients, rather than just performing well on historical or internal data.
- Context-Dependence: The authors considered the possibility that treatment outcomes are highly context-dependent, assuming that subtle differences in trial protocols, inclusion criteria, or recruitment sites (which cannot be modeled as predictors within a single trial) significantly impact a model's performance.

Methodology

The research utilized a rigorous, multi-scenario validation framework to test the generalizability of machine learning models in predicting treatment outcomes for schizophrenia. The methodology is structured around five international clinical trials and four distinct validation approaches to assess model performance.

Data Sources and Study Population

The researchers used data from five international, multisite randomized controlled trials (RCTs) obtained through the Yale Open Data Access (YODA) Project.

- Participants: The trials included 1,962 total patients (reduced to 1,513 for the final analysis based on data availability) aged 12 to 81.
- Diversity: The samples represented a heterogeneous population including teens, first-episode adults, chronic adults, and older adults across four continents.

Predictors and Outcome Measures

The models utilized 217 predictor variables extracted from baseline patient information.

- Predictor Types: These included demographics, psychiatric history (e.g., age of diagnosis, hospitalisations), clinical scales (PANSS, CGI), biometric data (blood chemistry, hematology, urinalysis), and treatment randomization.
- Dose Standardisation: Antipsychotic drug doses were standardised to paliperidone dose equivalents using the defined daily dose method.
- Primary Outcome: The Remission in Schizophrenia Working Group (RSWG) criteria at a 4-week timepoint was the primary metric. For sensitivity analysis, they also tested 25% and 50% symptom reduction and baseline-adjusted percent change.

Machine Learning Framework

- Algorithm: The primary method was the elastic net algorithm, a penalized regression method that handles correlated covariates and sparse predictors. Random forest models were also used for sensitivity analyses to detect complex non-linear patterns.
- Hyperparameter Tuning: Researchers performed a grid search across 400 combinations of penalty parameters (alpha and lambda) using repeated 10-fold cross-validation.
- Missing Data: Variables with more than 20% missingness were dropped. Remaining missing data were imputed using median values, assuming they were missing completely at random.

Validation Scenarios

The core of the methodology involved evaluating models across four distinct scenarios to measure true generalizability:

1. Within-trial (No Validation): Training and testing on the same sample to establish a performance baseline (risking overfitting).
2. Within-trial (Cross-Validation): Splitting a single trial's data into 10 subsets to estimate internal accuracy.

3. Paired-trial Validation: Training a model on one independent trial and testing it on another.
4. Leave-one-trial-out Validation: Training on four aggregated trials and testing on the excluded fifth trial to see if more data improved performance.

Statistical Evaluation

Model performance was primarily reported using balanced accuracy - calculated as $(\text{sensitivity} + \text{specificity}) / 2$ - because its null distribution is centered at 50%. The researchers also used bootstrapping to determine if accuracies exceeded chance and applied the Benjamini-Hochberg adjustment to account for multiple comparisons.

Main Findings

The research indicates that the generalisability of machine learning models for predicting clinical outcomes is largely "illusory". While models appear highly successful within the specific context in which they were developed, they fail to maintain that accuracy when applied to independent clinical trials.

The main findings include:

- High Internal but Low External Performance: Models predicted patient outcomes with high accuracy within the trial in which they were developed, achieving an average balanced accuracy of 0.72. However, when applied to independent, out-of-sample trials, the models performed no better than random chance, with average accuracy dropping to 0.54.
- Failure of Data Pooling: Aggregating data across multiple trials to train a model (the leave-one-trial-out approach) did not improve generalisability. This suggests that simply having more data from different sources is not enough to overcome the context-dependence of the predictions.
- Inadequacy of Internal Cross-Validation: The study found that standard within-trial cross-validation - a common practice in the field - yielded performance estimates (average 0.60) that were still significantly higher than what the models actually achieved on truly independent patients.
- Consistency Across Metrics and Algorithms: These results were not limited to a specific definition of treatment success or a single machine learning method. The same pattern of poor generalisability was observed across four different clinical outcome definitions and when using both elastic net and random forest algorithms.
- Identification of Key Risks: The researchers attributed the failure of these models to two primary factors:
 - Overfitting: Models may be fitting the "random noise" of a specific dataset rather than a true biological signal.
 - Poor Transportability: Models lack external validity because patient populations, recruitment procedures, and trial protocols differ in subtly important ways across clinical settings.

The authors conclude that current models for predicting treatment outcomes in schizophrenia are fragile and that excellent performance in one context is not a reliable indicator of success for future patients in different settings.

Limitations

The research identifies several significant limitations regarding the current ability of machine learning to predict treatment outcomes, primarily centered on the illusory nature of model generalizability.

Data and Methodological Limitations

- Missing Long-Term Data: Researchers originally intended to examine both early and late treatment responses, but "very high levels of missingness" in the long-term follow-up data forced them to focus exclusively on early treatment response at the 4-week mark.
- Imputation Effects: While missing data was considered to be missing at random, the use of median value imputation to preserve observations may have "somewhat reduced variability" within the dataset.

- **Missing Variable Types:** The models relied on clinical, sociodemographic, and simple biomarker data but did not include psychosocial information, social determinants of health, or neuroimaging and genetic data, all of which might be relevant for more accurate predictions.
- **Limited Diversity in Outcomes:** Although the results were consistent across four outcome definitions, they were all transformations of the same PANSS scale, which may not capture all facets of patient recovery.

Clinical and Biological Limitations

- **Diagnostic Heterogeneity:** Schizophrenia is an "umbrella category" that groups together patients at vastly different disease stages. This clinical heterogeneity makes it difficult to find a universal predictive signal.
- **Treatment Mechanism:** The research notes that the mechanism of the drugs used (D2 dopamine receptor blockade) may be "too far downstream" from the primary pathology of schizophrenia or the symptom severity criteria used to measure success.
- **Inherent Predictability Limits:** For some patients, the choice of treatment may have no impact on their clinical course, representing an inherent limitation to the potential for any model to predict outcomes.

Statistical and Modeling Limitations

- **Overfitting and Context-Dependence:** Models often fit the "random noise" of a specific dataset rather than a true biological signal. Furthermore, models are highly context-dependent, meaning they are influenced by trial-specific factors (like recruitment procedures or inclusion criteria) that cannot be modeled as predictors because they do not vary within a single trial.
- **Inaccuracy of Internal Validation:** The study highlights that standard cross-validation - the most common method for testing models - still provides an "overly optimistic" and fundamentally limited insight into how a model will perform on truly unseen patients.
- **Algorithm-Specific Risks:** In sensitivity analyses, random forest models demonstrated "excessive overfitting" in training data compared to the elastic net algorithm, although both ultimately failed to generalize to independent trials.

Implications for practice

Redefining Model Validation

Practitioners are urged to treat cross-validation as an initial diagnostic tool rather than proof of readiness. The research indicates that models appearing accurate during development often perform near random chance when applied to independent datasets. Consequently, external validation is mandatory before deployment, involving testing against independent cohorts, different geographic or institutional settings, and future temporal data.

Data Quality and Generalisation

A key finding is that larger datasets do not automatically improve generalisation. Generalisation depends more on data relevance and consistency than on sheer volume. Before merging datasets, practitioners must investigate differences in recruitment procedures, population characteristics, and data collection methods to avoid creating models that fail in the field.

Distinguishing Failure Modes

The sources identify two primary reasons for model failure:

- **Overfitting:** When a model learns patterns specific only to its training data. This can be mitigated through regularisation, feature selection, and simpler architectures.
- **Transportability Failure:** A structural issue where relationships learned in one context do not hold in another due to environmental or population differences. This requires cross-site validation and domain knowledge integration to resolve.

Strategic Design and Predictor Selection

To ensure robustness, predictor selection should prioritize stable, theoretically meaningful variables over those tied to specific local workflows or administrative procedures. Evaluation frameworks should simulate real-world conditions by using strategies like "leave-one-location-out" training or testing historical data against future data.

Post-Deployment and Transparency

Model development is an ongoing process that continues after deployment. Essential practices include:

- Continuous monitoring for data drift and performance degradation.
- Periodic recalibration and scheduled retraining.
- Transparent communication regarding model limitations, confidence intervals, and failure conditions to maintain stakeholder trust.

Ultimately, the goal of predictive modelling must shift from achieving maximum accuracy on a single dataset to ensuring reliability, stability, and reproducibility across changing environments.